



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project:	Enclave; Cowart Residential
Location:	N/A
Description:	Carrington - A/B/C, Slab

Date:	November 27, 2024	
By:	JTH / TJH	
House Faces:	Worst	NorthEast

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	1,818 ft ²
Volume	15,829 ft ³
Percentage Glass	9 %
Number of Bedrooms	3

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	7867
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.056	Attic	2x6	R-18 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.077	Garage Door		R-13	
Roofs	#	U-Value	Description/Notes			
	R1	0.033	Flat Ceiling	2x4	R-30 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed	2x4	R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.34	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		22917 Btu/hr		25849 Btu/hr	
Minimum Nominal Capacity		2.55 Tons		7.57 kW	
Load Density		714 SF/Ton		4.2 W/SF	
Floor Area Served		1818.0 SF		1818.0 SF	
Mechanical Ventilation		N/A CFM		N/A CFM	
System Size		2.5 Tons		10.0 kW	
Leaving Air Temperature		60.6 deg F		96.3 deg F	
Installed Capacity		727 SF/Ton		5.5 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	37	69	53	
Loft	57	38	47	
Owner Suite	140	195	168	
Owner WIC	16	17	17	
Owner Bath	45	62	54	
Bath	18	26	22	
Bed 2	81	81	81	
Bed 3	80	77	79	
Hall & Closet	17	14	16	
Laundry	43	8	26	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

1422

RC Heating kWh 1990

Total kWh	3412
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SEER 14

